SEGALL: A Unified Active Learning Framework for Wireless Sensing Data Segmentation

NAIYU ZHENG, City University of Hong Kong, Hong Kong SAR, China
RUOFENG LIU, Michigan State University, USA
XIAOYI FAN, Jiangxing Intelligence Inc., China
CONG ZHANG, Guangdong Laboratory of Artificial Intelligence and Digital Economy (SZ) and Jiangxing Intelligence Inc., China
LEI ZHANG, Shenzhen University, China
ZHIMENG YIN*, City University of Hong Kong, Hong Kong SAR, China

Wireless sensing has emerged as a promising technology due to its inherently privacy-preserving and contactless characteristics, enabling a wide range of applications. Until now, a major challenge in translating research into real-world applications is achieving accurate data segmentation. This process involves identifying the start and end points of target activities within complex and extended time-series sensing data, forming the foundation for subsequent inference tasks. However, existing segmentation techniques in wireless sensing are often constrained to a specific signal type and exhibit limited performance in distinguishing fine-grained activities, particularly when faced with similar background signals. To address these issues, we present SEGALL, a unified segmentation framework capable of processing diverse signals. By leveraging a lightweight, signal-independent deep learning architecture coupled with active learning, SEGALL enables accurate segmentation for fine-grained activities, even under interference from activities with similar characteristics. Experimental evaluations on both synthetic and real-world datasets demonstrate that SEGALL significantly improves segmentation reliability across various modalities, including IMU, Wi-Fi, and mmWave signals. For example, it achieves a segmentation accuracy of 91.12% for mmWave signals, outperforming previous segmentation solutions that achieved 70.50%. Furthermore, SEGALL reduces manual labeling effort by 40.15% and maintains a lightweight computational cost, making it suitable for deployment on edge devices such as the Raspberry Pi.

CCS Concepts: • **Human-centered computing** → *Ubiquitous and mobile computing*.

Additional Key Words and Phrases: Wireless Sensing, Sensing Segmentation, Active Learning, Artificial Intelligence

ACM Reference Format:
Naiyu Zheng, Ruofeng Liu, Xiaoyi Fan, Cong Zhang, Lei Zhang, and Zhimeng Yin. 2025. SEGALL: A Unified Active Learning Framework for Wireless Sensing Data Segmentation. *Proc. ACM Interact. Mob. Wearable Ubiquitous Technol.* 9, 3, Article 156 (September 2025), 26 pages. https://doi.org/10.1145/3749492

*Corresponding author.

Authors' addresses: Naiyu Zheng, City University of Hong Kong, Hong Kong SAR, China, naiyzheng2-c@my.cityu.edu.hk; Ruofeng Liu, Michigan State University, Michigan, USA, liuruofe@msu.edu; Xiaoyi Fan, Jiangxing Intelligence Inc., Shenzhen, China, xiaoyifan@jiangxingai.com; Cong Zhang, Guangdong Laboratory of Artificial Intelligence and Digital Economy (SZ) and Jiangxing Intelligence Inc., Shenzhen, China, zhangcong@gml.ac.cn; Lei Zhang, Shenzhen University, Shenzhen, China, leizhang@szu.edu.cn; Zhimeng Yin, City University of Hong Kong, Hong Kong SAR, China, zhimeyin@cityu.edu.hk.

Permission to make digital or hard copies of all or part of this work for personal or classroom use is granted without fee provided that copies are not made or distributed for profit or commercial advantage and that copies bear this notice and the full citation on the first page. Copyrights for components of this work owned by others than the author(s) must be honored. Abstracting with credit is permitted. To copy otherwise, or republish, to post on servers or to redistribute to lists, requires prior specific permission and/or a fee. Request permissions from permissions@acm.org.
© 2025 Copyright held by the owner/author(s). Publication rights licensed to ACM.
2474-9567/2025/9-ART156 $15.00
https://doi.org/10.1145/3749492

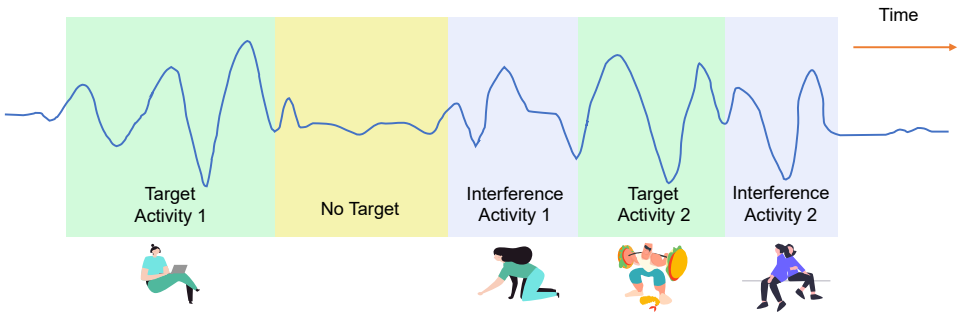


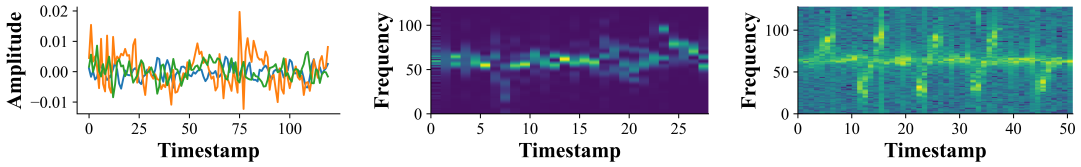
Fig. 1. The demonstration of wireless sensing system in real scenarios, which needs to segment target activities accurately from all activities.

1 INTRODUCTION

Wireless sensing has become a popular topics in recently years due to its advantages of privacy concern free and contactless. It is regarded as a promising technology to achieve ubiquitous perception. Numerous applications based on wireless sensing have gradually been adopted into practice use, such as gesture recognition [1, 9, 10], sleep monitoring [22, 26], fall detection [38, 50, 53], indoor localization [14, 17], etc. Recent advancements in AI (artificial intelligence) has pushed it to the higher level. Its ability of extracting complex features significantly simplifies the processing of various signals, *i.e*. Wi-Fi, mmWave, IMU, compared with traditional processing.

Although wireless sensing technology holds promise as a compelling alternative to cameras, a significant gap persists between research and real world. One of the primary challenges is the accurate segmentation of sensing data, which refers to identify the start and end points (boundaries) of each target activity. Most works [7, 57, 58] in research area assume that target data is pre-segmented, which means they collect each target activity data separately. However, in smart home scenarios, sensing data consists of long time series containing numerous target activities collected by Internet of Things (IoT) devices. Therefore, a critical step before recognition is accurate segmentation. Unlike fields such as computer vision (CV) [39, 45], which have developed advanced segmentation techniques to process continuous videos, the sensing field faces considerable challenges due to the unique characteristics of sensing data.

Previous researchers have explored real-time sensing signal-specific segmentation through various methods. The fundamental approach involves applying a sliding window with a fixed length [15]. Within each window, features (*i.e*. phase, amplitude) are extracted, and a specific threshold is set. If the average peak value exceeds this threshold, the period is classified as a segmentation. This method may be effective for simple tasks, such as fall detection. For instance, if a person suddenly falls while walking, the speed of the fall is greater than that of walking, allowing for easy detection using the aforementioned methods. When addressing fine-grained tasks such as gesture control, significant challenges arise due to the similarity in speed changes between gestures, which makes them difficult to distinguish. To overcome this issue, prior approaches [9, 28] often rely on pauses between gestures, serving as physical markers to identify the start and end of activity segments. However, in real smart home scenarios, it is unrealistic to require users to pause and only allow them to perform the target activities. Users engage in other activities, such as drinking water or using a smartphone, while attempting to control IoT devices with target gestures. As shown in Fig. 1, unlike controlled laboratory data collection, target activities and interference activities often overlap or interfere with each other, presenting additional challenges for sensing applications in real-world scenarios.



(a) IMU signal. (b) Wi-Fi signal. (c) mmWave signal.

Fig. 2. The spectrograms for three different kinds of signals.

Combined with aforementioned drawbacks, *is it possible to design a unified framework that can segment data across different signals under interference situations?* An intuitive solution is to apply a AI technology to the sensing field to achieve automated segmentation. However, designing such a unified architecture is not trivial. Specifically, there are following challenges that motivate us to design SEGALL. The first challenge is **real-time accurate segmentation under interference**, which refers to the model's acceptable performance in simple scenarios but significant degradation when fine-grained target activities are mixed with everyday activities as interference. Furthermore, the model is expected to lightweight enough to meet the real-time requirements. The second challenge is **complexity of different signals**, which arises from the diversity and heterogeneity of signal characteristics. Different signals have distinct physical properties, sampling rates, and noise distributions, as shown in Fig. 2. As the bandwidth and resolution of sensors continue to improve, the collected data becomes larger and more complex. The third challenge is **consuming segmentation labeling efforts**. Although automatic segmentation significantly alleviates the labeling burden compared to traditional iterative data collection, manually identifying and labeling each segmentation boundaries as ground truth is extremely time-consuming. This tedious procedure limits the complexity of data acquisition, which further constrains the development of wireless sensing. The fourth challenge is **scalability of the system downstream tasks**. The ultimate goal of segmentation is to serve various downstream tasks for recognition, such as gesture recognition and vital sign monitoring. However, it is not flexible if the system output segmentation and recognition simultaneously as the downstream requirements may change, which needs to redesign and retrain the model. It is worthy to explore whether the system can increase the scalability of downstream tasks without excessive additional resources.

To this end, we introduce SEGALL, a real-time active learning-based sensing data segmentation system designed to benefit IoT applications. SEGALL consists of three main components: a tailored lightweight deep learning model that is versatile across various signals, an active learning-based training framework, and an scalable integrated training process. To ensure real-time segmentation and address the complexity of heterogeneous signals, we propose a lightweight Transformer-LSTM-based architecture capable of processing signal inputs with varying dimensions and lengths. To alleviate the challenges associated with labor-intensive segmentation labeling, we integrate active learning into the training process and design a novel query metric to select the most informative samples for labeling, thereby reducing labeling efforts. Moreover, the proposed deep learning architecture, combined with active learning, effectively mitigates interference challenges. Specifically, the model formulates the task as a binary classification problem, simplifying the recognition of target activities while suppressing interference. The LSTM layer further refines the sequence processed by the Transformer, filtering out interference activities. Additionally, the active learning query strategy quantifies the degree of disorder in long-term sensing sequences, enabling the identification of highly interfered data for labeling and training. To enhance the scalability of downstream tasks, we design an integrated training approach that leverages the encoder trained during the segmentation phase for downstream tasks. The system reuses parts of the existing architecture and modifies only the task-specific components, ensuring scalability without requiring significant additional resources. In general, SEGALL operates in two key phases.

During the upstream segmentation phase, the original long time series data is fed into a Transformer-LSTM-based architecture, which is empowered by active learning to segment the data into single activity segment. In the downstream task phase, the segmented data from segmentation phase is processed through the same encoder, and a task-specific classifier outputs the final predictions (for example, in gesture recognition).

We evaluate SEGALL on both a self-custom dataset collected from real smart home scenarios and synthetic datasets. The results show that SEGALL can achieve more than 94.64% accuracy across three different modality signals. Empowered by active learning, the segmentation labeling is reduced by at least 38.17%. Proposed integrated training process not only increases the scalability of downstream tasks but also accelerates the training by 75%. We also measure the real-time ability of our system on devices with different architectures. Note that we select IMU, Wi-Fi, and mmWave as demonstrations in evaluation part. Although various signals differ in physical properties and pre-processing methods, they can be transformed into multivariate time series after appropriate signal-specific processing. SEGALL operates on this shared representation, applying a unified segmentation model that focuses on temporal dynamics rather than modality-specific features. This abstraction not only enables cross-modal applicability but also allows easy extension to other sensing types, as long as their outputs can be represented as temporal sequences.

In summary, we make the following technical contributions in this paper:

(1) To our best knowledge, SEGALL is the first unified framework which focuses on segmenting sensing data across various signals under interference. We believe the automation of SEGALL can help not only researchers but also benefit the IoT product commercialization in smart-home scenarios.
(2) We design a tailored learning-based segmentation model which can also be reused in downstream tasks, increasing the scalability and reducing the overall training cost greatly. The lightweight model is able to achieve real-time sensing.
(3) We introduce active learning with a novel query strategy into sensing area as SEGALL core training workflow. It further picks the higher valuable samples and reduces the segmentation labeling efforts.
(4) We demonstrate the good performance of our approach through experiments on both custom datasets and synthetic datasets across various signals.

In the following of the paper, Section 2 introduces related work. Section 3 demonstrates our motivational study and the detailed design of SEGALL is described in Section 4. The experimental results of the evaluation are presented in Section 5. Discussion and conclusion are reported in Section 6 and 7 respectively.

2 RELATED WORK

In this section, We first introduce background of segmentation and active learning in sensing field to better understand the limitations of existing work.

2.1 Wireless Sensing

Wireless sensing primarily focuses on extracting and analyzing features from raw wireless signals. One widely studied area is WiFi-based sensing, which has shown great potential in applications such as human activity recognition, localization, and gesture detection [11, 20, 32, 40]. These systems typically utilize Channel State Information (CSI) [41, 52] or Received Signal Strength Indicator (RSSI) [34, 47] to infer physical situations. Beyond WiFi, other technologies such as millimeter-wave (mmWave) radar, inertial measurement units (IMU), and Long Range (LoRa) communications have been explored for sensing applications. mmWave radar systems offer higher resolution and robustness compared to WiFi due to their wider bandwidth [5, 25, 46, 55]. IMU-based sensing relies on accelerometers and gyroscopes to monitor physical activities, often combined with RF signals to enable hybrid systems that achieve higher accuracy [4, 44]. Meanwhile, LoRa-based sensing, which operates

on low-power, long-range communication channels, has shown promise in outdoor and large-scale sensing tasks, including environmental monitoring and object tracking [6, 43, 54].

More recently, generative AI techniques have opened new avenues for wireless sensing research. Generative models such as Generative Adversarial Networks (GANs) and Diffusion Model have been employed to augment training data, synthesize realistic signal patterns, and denoise raw signals [7], which helps improve the robustness of sensing systems.

2.2 Data Segmentation

Data segmentation plays a crucial role in interpreting continuous data streams by dividing them into meaningful segments for downstream tasks such as activity recognition and environmental monitoring. This task has been extensively studied in various fields, particularly in computer vision (CV), and has recently gained significant attention in the wireless sensing domain. In computer vision, segmentation refers to partitioning images or videos into regions with semantic or instance-level meaning [18, 19, 33], which is slightly different definition from sensing segmentation. Popular deep learning-based architectures, such as Fully Convolutional Networks (FCNs) [21, 27], have revolutionized segmentation by enabling pixel-level predictions with high accuracy.

In wireless sensing, segmentation focuses on extracting meaningful segments from continuous signal data. Unlike visual data, sensing data is often characterized by irregular patterns, environmental dependencies, and significant noise, making segmentation more challenging. Current mainstream approaches relied on threshold-based techniques, where changes in signal amplitude, phase, or other features are used to identify event boundaries [13, 15, 49]. For instance, Keenan et al. [15] proposes a threshold-based sliding window method based on CSI phase difference of Wi-Fi for fall detection. The process is computed as $\mu + 6\sigma \leq \tau$, where τ is the threshold value, μ is the mean, and σ is the standard deviation of the CSI phase difference. The system checks whether the signal transitions between 'fluctuating' and 'stable' states based on whether the standard deviation in the sliding window is above or below the threshold. Similarly, DeFall [13] utilizes SLN-DTW [30] to compare the real-time CSI-based speed and acceleration sequences with pre-generated fall template. If the DTW distance is below a pre-defined threshold, a fall is detected. Change point detection (CPD) techniques are regarded as another effective segmentation method, which refers to the process of identifying boundary points in time series sensing data where the statistical properties change significantly [2, 3, 31]. For instance, the method proposed in [3] employs a non-parametric sequential change detection approach, where a change score is computed between consecutive change point windows to quantify the degree of change. However, most CPD approaches rely on comparing change scores against a predefined threshold to detect change points.

Although the above methods have shown promising performance, sensing signals vary across different situations, which leads to fluctuations in the threshold values. For large-scale activities such as fall detection, it is relatively easy to identify the transition from normal walking to a sudden fall, as it results in drastic changes in feature values. In contrast, for fine-grained activities like gesture recognition, threshold-based and CPD-based techniques are easy to fail when the gestures are consecutively performed without enough pause. RFWash [16] proposes a monitoring system for hand hygiene that can be trained on unsegmented naturally-performed gesture sequences, but it mainly focuses on the accuracy of the hand movement sequence rather than each specific target gesture. WiCGesture [8] innovatively proposes a meta-motion segmentation approach without requiring explicit pauses, which decomposes continuous hand motion into basic geometric units termed "meta motions". However, it relies on predefined meta-motion sequences as templates for gesture matching. The predefined reference set is difficult to cover all possible variations when it involves completely random gesture sequences.

In more practical scenarios, target activities are often mixed with similar daily activities. Additionally, the feature pattern changes with different signals vary due to differences in resolution. This issue will be discussed further in

the motivation section. In general, data segmentation for sensing signals remains a challenging problem due to the complexity of wireless signals and interference.

2.3 Active Learning

Active learning [35, 37, 51] becomes a machine learning paradigm that focuses on selectively choosing and labeling the most informative and valuable instances from an unlabeled dataset to enhance the accuracy of a predictive model, instead of randomly sampling data or labeling the entire dataset. The original goal of active learning is to minimize the amount of labeled data needed to reach a desired level of model performance.

Various strategies have been proposed to determine which samples to query: Uncertainty Sampling [48], Query by Committee [12], Diversity Sampling [23], etc. The performance of AL-based methods heavily depends on the selection of query strategies. However, the effectiveness of these strategies often varies across specific tasks. Most previous works have focused on hard classification problems, where the goal is to identify a single sample rather than multiple segments within a long time series. Recent research [29] indicates that no active learning method consistently outperforms other approaches. This suggests that designing suitable active learning algorithms tailored to specific scenarios is crucial.

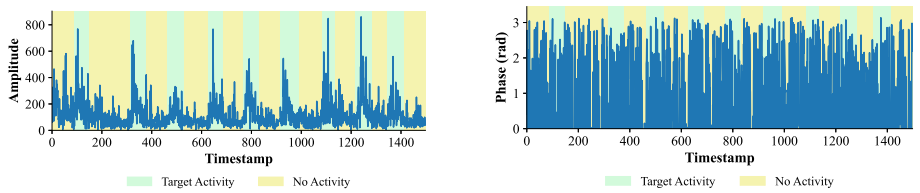
In sensing domain, few studies have explored active learning, as it is still a novel concept in this field. The existing work [56] applies active learning only to human activity recognition (HAR), which fundamentally remains a hard classification problem. In contrast, segmentation problems are more challenging and better aligned with practical applications. Based on the original goal of active learning, the query strategy needs to select target activities and exclude interference activities in our scenarios. This inspires us to design a new active learning algorithm that not only focuses on reducing labeling efforts but also resists interference.

3 MOTIVATION

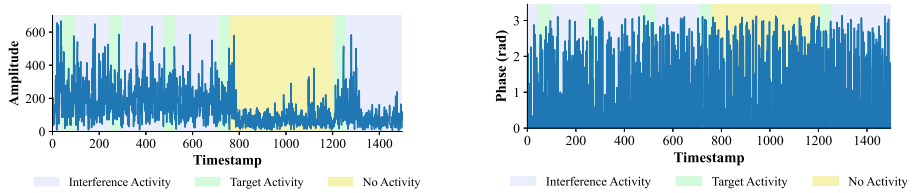
To motivate the design of SEGALL, we implement the motivational study in this section.

Practical Scenarios under Interference. An intuitive segmentation method involves setting an empirical threshold for the start and end points of each activity. Based on this approach, researchers utilize different features for detection (*i.e.* , amplitude, phase), as discussed in Sec. 2. These methods are effective when the target segmentation is straightforward. The simplicity here has two aspects. On the one hand, they are more suitable for coarse-grained activities, such as walking and running, which generally exhibit distinct patterns as amplitude and speed change. On the other hand, while some methods are applicable to fine-grained activities, such as gesture recognition or respiratory detection, they necessitate a relatively pure data collection process.

We conduct several experiments to present above limitations. We first ask volunteers do target gestures without extra interference. It can be clearly observed in amplitude-timestamp relation, as shown in Fig. 3a, which peaks are corresponding to the target activities. However, if we set a more complex and practical scenario under interference condition, it is not intuitive anymore. Specifically, except for the target activities, the volunteer continuously does daily activities (interference), such as typing the keyboard or grabbing a glass for drinking. The amplitude values between target and interference activities are completely mixed with each other in Fig. 3a. As for phase feature, it is widely used in previous Wi-Fi sensing works. However, it fails in mmWave sensing. Compared with Wi-Fi (2.4 GHz or 5 GH), mmWave operates at higher frequencies (24 GHz, 60 GHz, or 77 GHz), resulting in a shorter wavelength and much higher spatial resolution. High spatial resolution causes the target (e.g., a gesture or movement) to span across multiple range bins. Because of this, the phase in each range bin will vary significantly and wrap around (phase ambiguity). For respiratory detection, the fluctuation of the chest cavity is small within one range bin, which can be detected by phase change. But for gesture recognition, the phase change is no longer consistent across range bins, making it unreliable for sensing, as shown in Fig. 3b and 3d.



(a) Amplitude without interference activities. (b) Phase without interference activities.



(c) Amplitude with interference activities. (d) Phase with interference activities.

Fig. 3. Amplitude and phase change of mmWave signal under clean and interference scenarios.

We further measure the performance of threshold-based segmentation methods [9, 15], as displayed in Table. 1. Consistent with Fig. 3, all the baselines only work with amplitude under ideal (non-interference) scenario but fail in practical scenario, regardless of which features are used. This reminds us to consider the differences caused by signal resolution and underlying characteristics, which bring more challenges to design a unified framework. Moreover, it demonstrates that simply using single dimension feature lacks sufficient representational ability.

Table 1. The segmentation ability of the threshold-based model under ideal and practical scenarios.

Scenario	Ideal	Practical
Amplitude	√	X
Phase	X	X

The complexity of different signals. The target of interest evolves over time, and each signal exhibits distinct features at each time point. The complexity of this representation becomes particularly evident for signals with multiple carriers, such as Wi-Fi and mmWave. For example, an IMU (Inertial Measurement Unit) produces a relatively simple signal with three dimensions, corresponding to measurements along the x, y, and z axes. In contrast, the DFS (Doppler Frequency Spectrum) of mmWave signals can have up to 128 dimensions, representing the strength of individual frequency components. This highlights the significantly higher complexity of mmWave signals compared to IMU signals.

Table 2. The segmentation accuracy of the basic model trained with different signals.

Signal Type	IMU	Wi-Fi	mmWave
Accuracy	82.17%	12.33%	14.30%

To evaluate the impact of signal complexity, we conduct experiments using long time-series data from IMU, Wi-Fi, and mmWave. We train a standard LSTM-based model (two stacked LSTM layers)—a widely used baseline

for time-series data—separately on each signal type. As shown in Table 2, the model achieves significantly better performance on IMU data, with an average accuracy of 82.17%, compared to just 12.33% for Wi-Fi and 14.30% for mmWave. These results suggest that simple models like vanilla LSTM struggle to effectively capture the high-level features required for more complex signals like Wi-Fi and mmWave. This disparity underscores the importance of carefully considering signal complexity when designing unified models for multiple wireless signals. Further challenge based on this lies in balancing model scale and resource cost. Simply increasing the number of trainable parameters to handle complex signals may lead to overfitting on simpler signals, such as IMU. Additionally, larger models often fail to meet real-time performance requirements due to their computational overhead. On the other hand, overly simplistic models, like the vanilla LSTM, are insufficient for extracting meaningful features from high-dimensional, complex signals such as mmWave. Therefore, developing a balanced model that can generalize well across signals of varying complexity, while maintaining efficiency for real-time deployment, is crucial.

Segmentation Labeling Efforts. In practical scenarios, the distribution of time periods for performing the target activities is very random, rather than performing continuously in ideal settings. This is illustrated in Fig. 4, where the vertical red lines indicate specific time points of target activities, and the white boxes represent random daily activities that can be considered as interference. Due to the sampling rate (*i.e.* 50 Hz for mmWave setting) of data collection, one minute can correspond to 3000 timestamps. Although the total number of timestamps is reduced to 1000 in the figure after downsampling, it is time-consuming to accurately find boundaries of each target activity to label ($\frac{10}{1000}$ as each boundary is less than 0.1 second, as five activities in Fig. 4). Moreover, the intervals between target activities are irregular, which could be consecutive in extreme situations (the last two activities in the figure). This brings more challenges for segmentation labeling.

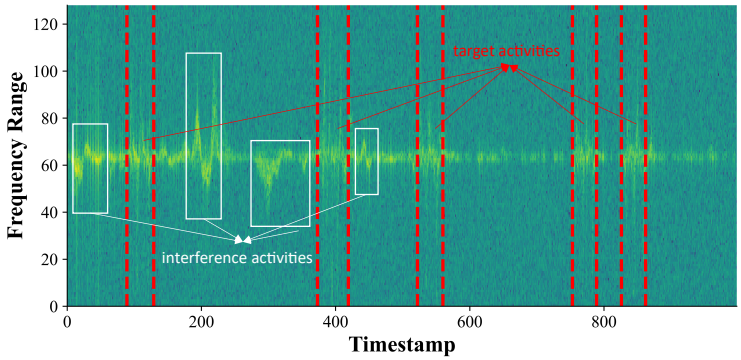


Fig. 4. The spectrogram of collected mmWave data in one minute with target activities and interference activities.

Furthermore, it is simple for model to train the data if all gestures are performed in a standard manner. However, when different individuals perform the same gestures, the resulting patterns can vary significantly due to differences in arm length and speed. Even for the same person, there can be slight variations in gesture execution over time. Labeling a large amount of easily distinguishable segment can become repetitive and labor-intensive. Instead, it would be more beneficial to focus on labeling the most informative segment from unlabeled long time-series sensing data. Therefore, a better method is expected to be designed to reduce segmentation labeling efforts.

Increasing the scalability of system. The ultimate goal of segmentation is to serve a variety of downstream tasks. Additionally, user's demands of downstream tasks may change over the time. However, directly coupling segmentation and recognition within the same system limits flexibility, as it often necessitates redesigning and retraining the model for each new task. This process can be time-consuming and resource-intensive. While the final objectives of segmentation and downstream tasks differ, a shared aspect is the extraction of base features from

the original sensing data. Therefore, the partial structure trained during the segmentation phase has the potential to generalize and extract suitable features for downstream tasks. This insight opens the possibility of enhancing scalability by reusing the trained structure while modifying only specific components of the system. Specifically, instead of redesigning the entire model, it is promising to replace the segmentation classifier with task-specific components for different downstream applications. We hope this approach minimizes the need for additional resources while maintaining the system's adaptability to diverse tasks. These analyses motivate us to explore an integration method that improves the scalability and effectiveness of SEGALL, enabling it to support multiple downstream tasks.

4 OUR APPROACH: SEGALL

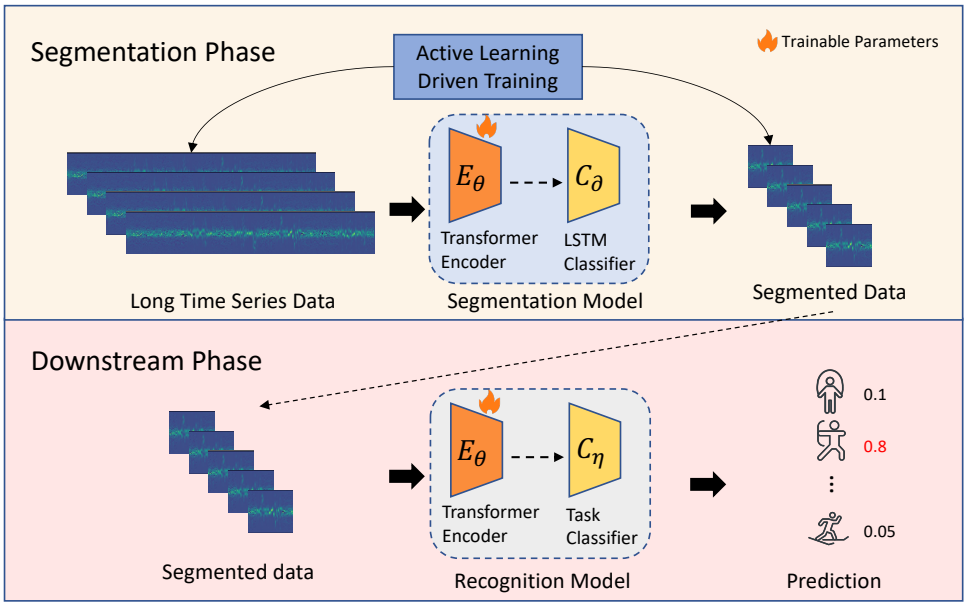


Fig. 5. Workflow of SEGALL.

Based on the discussions presented in Section 3, we hereby propose our solution, named SEGALL, which utilizes active learning to automatically segment continuous long-time series sensing data by lower cost and is able to adapt to downstream tasks. SEGALL comprises three novel designs: signal-independent segmentation architecture, active empowered segmentation, and scalable integration. We first describe the overview of SEGALL and then introduce our three novel designs in sequence.

4.1 Overview

The workflow of our approach is shown in Fig 5. SEGALL mainly contains two stages, segmentation phase and downstream task phase. Here we take gesture recognition as one of challenging downstream tasks which needs to distinguish different fine-grained actions. Therefore, SEGALL needs to accurately segment data and can be adapted to various tasks flexibly. At segmentation stage, we implement a signal-independent segmentation architecture to segment data to receive series contains target activities, which is supposed to be effective for various signals. During this stage, SEGALL adopts active learning to empower training process to reduce segmentation labeling

efforts and resist interference. We design a new query strategy to select the most informative samples for annotation. Afterwards, the well-segmented data is sent to the downstream task stage for recognition. SEGALL is able to reuse the main architecture to accelerate the process and increase flexibility of downstream-task.

To resist the interference from human non-target activities, SEGALL leverages the self-attention mechanism in a Transformer-LSTM-based model to focus on key features of target gestures while suppressing interference. SEGALL also labels interference as background and applies Focal Loss to reduce bias toward dominant non-target activities. Additionally, the proposed active learning strategy prioritizes heavily interfered samples, improving the model's robustness through targeted learning.

As for preprocessing of wireless signals, due to the drawbacks discussed in Sec. 2, the DFS (Doppler frequency spectrum) is expected to alleviate the issue of insufficient representational ability, which transfers the signal to frequency domain. It focuses on motion-induced frequency shifts and reflects dynamic changes in the wireless environment, which provides a more comprehensive representation of motion, including velocity, direction, and potentially acceleration. We follow the standard procedures in prior literature to extract the DFS [58] from raw signals.

We will introduce the main components in more detail in the following subsections.

4.2 Signal-independent Segmentation Architecture

The model is supposed to handle distinct modality signals with various timestamps due to the uncertainty of target activity length. Furthermore, to ensure the real-time requirements, the model needs to be lightweight enough. Therefore, we design a generic neural network architecture for sensing, which consists of Transformer and LSTM layers, as shown in Fig 6.

The long time series data is first processed by Transformer-based encoder at the segmentation stage, which captures the basic sensing features. There are three key insights for SEGALL to use Transformer architecture rather than typical LSTM architecture for time-series data. First, as explained in Sec. 3, the time distributions of target and interference activity are very random. The self-attention mechanism of Transformer captures global dependencies regardless of temporal alignment by calculating attention weights between all time steps in the input sequence, which focuses more on target activities to resist interference. Second, Transformer allows for parallel processing of input data, making them ideal for signals with irregular timestamps. Furthermore, parallel processing speeds up training and inference times to meet real-time requirements. Last, the multi-head attention mechanism enhances feature extraction, which learns different representations of the input, focusing on distinct patterns and modalities from IMU to mmWave.

The high-level embedding is generated after Transformer-based encoder. As discussed in Sec. 3, the target activities can be very sparse (*i.e.* only five targets in one minute). Therefore, the key question is that how we define the ground truth of segmentation to differentiate the target and interference. Specifically, if the model only output the start and end points of each activity as a regression problem, the target label imbalance will seriously damage performance as the model is sensitive to activities whether it is from target or interference. As a result, the output will be wrong with random points in the whole sequence. We propose additional designs to address this issue in SEGALL. Unlike traditional regression problems, SEGALL treats this challenge as a binary classification problem. In our framework, the duration of the target activity is labeled as 1, while all other time intervals including interference are labeled as 0. This approach increases the proportion of the target label, thereby alleviating label imbalance. Furthermore, inspired by [24], we incorporate Focal Loss as a supplementary loss function. Focal Loss is particularly effective in emphasizing the learning process for misclassified or hard-to-classify examples, which enhances performance on minority target activities and mitigates bias towards the interference. Through the design of ground truth and loss function, the model focuses more on the target and tries to ignores the interference Before feeding the embeddings into the final binary classifier, we introduce an LSTM-based layer to enhance the

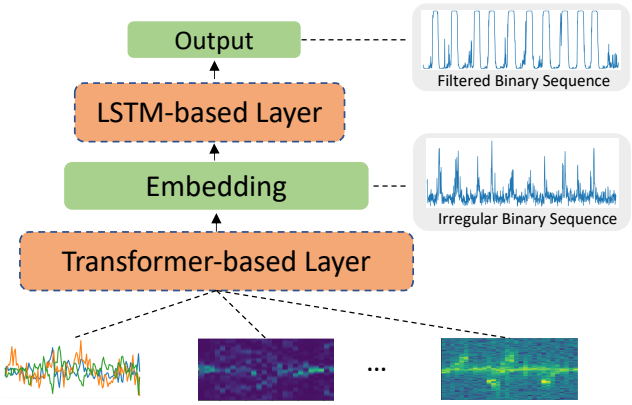


Fig. 6. The architecture of segmentation model.

representational capabilities of SEGALL. We found that relying solely on a Transformer layer is insufficient for compressing multi-frequency information into a single binary sequence. As shown in Fig. 6, the binary sequence embedding generated by the Transformer layer is full of noise and irregular. LSTM is particularly well-suited for modeling local sequential dependencies due to its recurrent structure and internal memory states. It complements the Transformer by filtering and smoothing the noisy binary embeddings through step-by-step sequential modeling. This not only enhances the temporal coherence of the feature representation but also helps the model better discriminate between sparse target activities and surrounding interference or noise. As a result, the LSTM layer produces a cleaner, more stable embedding sequence for the final binary classifier. This hybrid design leverages the global receptive field of Transformers and the local refinement capabilities of LSTM, improving both robustness and segmentation precision in real-world sensing scenarios.

At recognition stage, SEGALL share the same encoder with segmentation stage to improve the training efficiency and benefit from pretraining knowledge. The high level embedding is subsequently transmitted to task-specific module (here we take gesture prediction as an example). SEGALL aggregates embeddings in high dimensions and select the average observation result as the final output.

Suppose the input long time series data is denoted as $s \in S = \{s^1, s^2, ..., s^n\}$ at segmentation stage. The dimension of s is $t \times f$, where t is the length of time series data (timestamps), and f is the frequency range. For different signals, the frequency range might change. The data is first fed into the encoder E to generate embeddings emb, which can be represented as $emb = E(s, \theta)$, θ is trainable parameters for encoder E. The dimension now is $t \times hidden$, where $hidden$ is the the feature dimension of the hidden layers in Transformer to indicate the dimensionality of the feature representation for each time step. The high level embedding emb is further processed to binary sequence emb_{bi} through dimensionality reduction, consistent with ground truth. Then the LSTM-based binary classifier C_{seg} conducts $\hat{y}_{seg} = C_{seg}(emb_{bi}, \alpha)$, where α is parameters of classifier and $\hat{y}_{seg}$ is the predicted probability that the timestamp belongs to the positive class. We adopt Sigmoid function to transform it to a classification probability. The output can be represented as one-dimension sequence $\hat{y}_{seg} \in R^{t \times 1}$. For classification, SEGALL uses binary cross entropy loss:

$$L_{bce} = -\frac{1}{T}\sum_{i=1}^{T}\left(y^i_{seg}\log(\hat{y}^i_{seg}) + (1 - y^i_{seg})\log(1 - \hat{y}^i_{seg})\right) \tag{1}$$

where y^i_{seg} is the true label (0 or 1) for the ith timestamp and T is the total timestamps. As discussed above, we refer to Focal Loss to alleviate class imbalance, the original expression is:

$$L_{focal} = -\alpha_t(1-p_t)^\gamma \log(p_t) \tag{2}$$

where p_t is the predicted probability for the positive class, α_t is a weighting factor for the class (to balance the importance of positive/negative examples) and γ is the focusing parameter that adjusts the rate at which easy examples are down-weighted. We further generalize it to binary cross entropy to benefit from both of them:

$$\begin{aligned} L_{seg} = -\frac{1}{T}\sum_{i=1}^{T}\Big[&\alpha_t y^i_{seg}(1-\hat{y}^i_{seg})^\gamma \log(\hat{y}^i_{seg}) \\ &+ (1-\alpha_t)(1-y^i_{seg})(\hat{y}^i_{seg})^\gamma \log(1-\hat{y}^i_{seg})\Big] \end{aligned} \tag{3}$$

After completing the segmentation, the original long time series data s is segmented to numerous separate fragments contain target activities, donated as $x_i = x^1_i, x^2_i, ..., x^m_i$. As same as the first stage, x_i is processed by E to generate emb. However, due to the difference of various downstream tasks, it is necessary to modify the task specific module. Here we take gesture classification as an example. To ensure both of real-time requirements and recognition accuracy, we aggregate the observation results from all time steps $\frac{1}{T}\sum_{i=1}^{T} emb[i,:]$, then send the result to final linear classifier C_{task} to conduct the prediction $\hat{y}^i_{task} = C_{task}(emb, \eta)$, where η is the trainable parameters of classifier. Minimizing the following loss to optimize classification results:

$$L_{task} = \frac{1}{n}\sum_{i=1}^{n} L_{cross-entropy}(y^i_{task}, \hat{y}^i_{task}) \tag{4}$$

where $L_{cross-entropy}$ is the cross entropy loss and y^i_{task} is the label for the input x_i.

4.3 Active Learning Empowered Segmentation

Although the tailored architecture addresses the compatibility and alleviates interference while unifying various wireless signals, it still falls short of meeting realistic requirements. Specifically, when deploying SEGALL for smart home applications, the system is expected to achieve acceptable segmentation accuracy with minimal labeling efforts. In traditional methods, models are trained with pre-labeled dataset. While collecting data is straightforward—requiring only the activation of the device—labeling all the boundaries for segmentation can be both expensive and time-consuming. Furthermore, solely relying on the deep learning model to implicitly eliminate interfering activities is not enough as the output of model is black-box. Therefore, the system should prioritize focusing on data that is difficult to distinguish due to interference or the presence of actions unrelated to the target activities. To enhance the effectiveness of segmentation, we propose the use of active learning as a solution. Active learning allows the model to identify and label the most informative samples among interference, thereby optimizing the labeling process and improving overall model performance.

Overall Process. Active learning is a technique within machine learning that focuses on the iterative selection and annotation of the most informative instances from an unlabeled dataset to enhance the performance of predictive models. The fundamental principle of active learning lies in the strategic selection of examples for labeling, as opposed to random sampling or labeling the entire dataset. This method seeks to minimize the amount of data needed to reach a specified level of accuracy for the model.

Suppose we have collected a sensing dateset but have not labeled them yet, denote it as unlabeled data pool D. Each sample in the dataset is a long time series that contains target and interference activities. For the initial epoch of training, we need to randomly select a few samples and manually label them. This step is for assigning initial weights to the model while the model cannot make predictions or assess uncertainty in the remaining unlabeled data without any labeled data. These initial labeled data is removed from D to labeled data pool $\hat{D}$. Our proposed

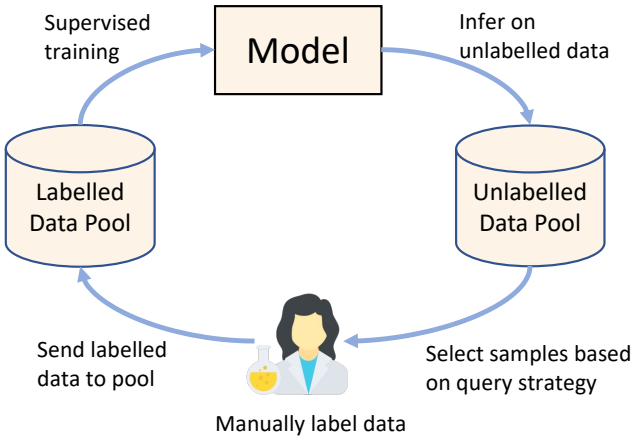


Fig. 7. The active learning architecture of SEGALL.

Transformer-LSTM-based model is trained by these labeled data. Then a query strategy Q is employed to identify the most informative instances from the remaining unlabeled data pool D. The objective is to select samples that are complex or uncertain for the current model, as these instances are expected to yield the most significant insights. In our scenarios, the selected samples are normally contains interference activities that are difficult for the model to distinguish with target activities. After selection, we continue to manually label them and put them into $\hat{D}$. The model utilizes all labeled data in $\hat{D}$ at this stage for training and updating its parameters. The last step is repeating data selection and model training until the model performance achieve an acceptable level. For the convenience, the core algorithm is summarized as Algorithm 1

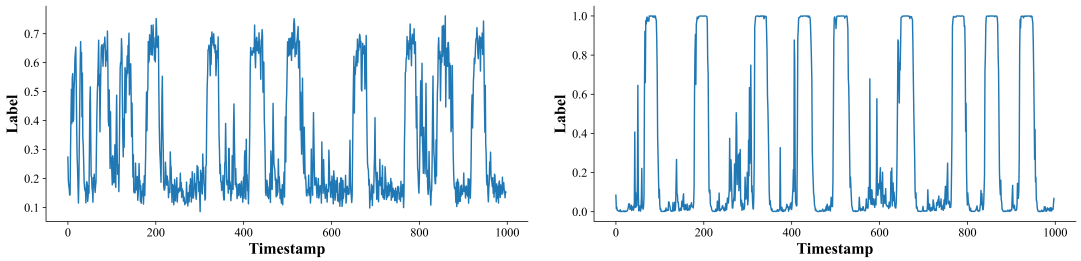
Algorithm 1 Active Learning Process

Initialization: Select an initial small random subset of the unlabeled dataset D and label it. Then add it into labeled data pool $\hat{D}$. Train the model M on the dataset $\hat{D}$.
While not reach maximum iteration T **do**
  Predict all data in D
  if not reach the maximum number of uncertain samples to be selected N **then**
    Use query strategy Q to select K uncertainty samples from D
    Manually annotate these samples and add them into $\hat{D}$
    Update M via fine-tuning with $\hat{D}$
  else
    break

Query Strategy. The query strategy is crucial in active learning because it directly influences the efficiency and effectiveness of the model's learning process by determining which unlabeled instances will be selected for annotation. By focusing on instances that the model is least confident about, or that are expected to provide the most significant improvement in predictive accuracy, the query strategy ensures that each labeling effort yields maximum benefit. We hope the targeted approach not only reduces the costs associated with manual labeling but also further enhances the ability to resist interference activities, as it prioritizes data points that are likely to enhance the model's understanding of the underlying patterns in the data.

There are several common strategies. **Uncertainty Sampling.** Selecting instances where the model is least confident in its predictions, often based on metrics like prediction probability or margin between the top two class probabilities. **Query by Committee.** Using multiple models (the committee) to assess uncertainty, where instances with the most disagreement among the models are chosen for labeling. **Expected Model Change.** Selecting samples that are expected to cause the most significant change in the model's parameters if labeled. **Expected Error Reduction.** Choosing instances that are predicted to lead to the greatest reduction in future error when added to the training set. **Diversity Sampling.**: Ensuring a diverse selection of instances to cover different aspects of the data distribution, which can improve generalization.

Due to the complexity interference of signals, simply relying on previous query methods may lead to inaccurate prediction as the most informative samples are not adequately selected. Specifically, uncertain sampling only focuses on the final prediction probability but ignores intermediate steps and features of input. Although committee query can aggregate the knowledge from different experts, it is not realistic to train multiple models as high efficiency requirements. Furthermore, the time and frequency of target activities are random in each period, as well as interference. Sampling based on data distribution is not suitable.



(a) Segmentation output disorderly series of model during the training process.

(b) Segmentation output tidy series of well-trained model.

Fig. 8. The comparison of segmentation phase series output during different processes.

The output of segmentation model is 0-1 time series, which 1 represents the period of target activities and 0 represents the spare time or interference. A phenomenon can be observed that the output time series for data contains more interference is haphazard as it is more difficult for model recognition, as shown in Fig 8. The model is more likely to recognize interference activities as target activities. In other words, dispersion is expected to measure how spread out the values are. Therefore, we use variance as basic metric,

$$\sigma^2 = \frac{1}{N}\sum_{i=1}^{N}(x_i - \mu)^2 \tag{5}$$

where N is the number of observations in the population, x_i represents each individual observation, and μ is the population mean. Considering the time and frequency of target activities are random in each data, as well as interference, a new variance based metric is designed. From Fig 8a, if the peak values are scattered more randomly (higher interference), the transitions between 1 and 0 will increase. The number of transitions ($\mathcal{M}$) is defined as the number of times the time series switches between 0 and 1 (or vice versa). For a sequence $x_1, x_2, \ldots, x_N$, it can be computed as:

$$\mathcal{M} = \sum_{i=2}^{N} |x_i - x_{i-1}| \tag{6}$$

This value directly reflects the randomness of the output: higher $\mathcal{M}$ corresponds to more frequent transitions and greater interference. Then we introduce a weighting factor based on $\mathcal{M}$ to adjust the variance. Since higher $\mathcal{M}$ indicates more randomness, the metric should increase with $\mathcal{M}$. The combined query metric is named as Dispersion Metric and defined as:

$$Q = \sigma^2 \cdot \left(1 + \frac{\mathcal{M}}{N-1}\right) \tag{7}$$

$\frac{\mathcal{M}}{N-1}$ normalizes the number of transitions by the maximum possible number of transitions $(N-1)$ for a time series of length N. $1 + \frac{\mathcal{M}}{N-1}$ serves as a scaling factor to amplify the standard variance (σ^2) based on the degree of randomness. SEGALL utilizes this query metric to further measure uncertainty and value of unlabeled data, thereby enhances the recognition ability of interference activities and reduces the labeling efforts.

4.4 Scalable Integration

To increase the scalability of SEGALL efficiently, we propose a modular design that leverages the encoder trained during the segmentation phase for downstream tasks. This design enhances scalability by allowing the partial reuse of the trained structure while incorporating task-specific components to meet diverse application needs. The encoder is designed to serve as a general-purpose feature extractor, decoupled from task-specific classifiers. During the segmentation phase, the encoder learns to extract base features from time series data derived from wireless signals. These features, which are robust across variations in signal type and sequence length, are expected to generalize well to downstream tasks. For downstream applications, instead of redesigning and retraining the entire model, the system retains the pre-trained encoder and integrates lightweight, task-specific modules. This modularity minimizes the computational cost and resource demands while maintaining flexibility to adapt to evolving user requirements.

The proposed design follows a two-stage process. Pretrained Encoder Reuse: The encoder, trained during segmentation, is partially or fully trainable depending on the complexity of the downstream task. This ensures that the foundational knowledge from segmentation is retained while allowing some adaptability to task-specific data. Task-Specific Module Design: Lightweight classifiers or other task-specific modules are introduced to replace the segmentation classifier. These modules are fine-tuned on downstream task data, leveraging the pretrained features from the encoder to accelerate convergence and improve task performance.

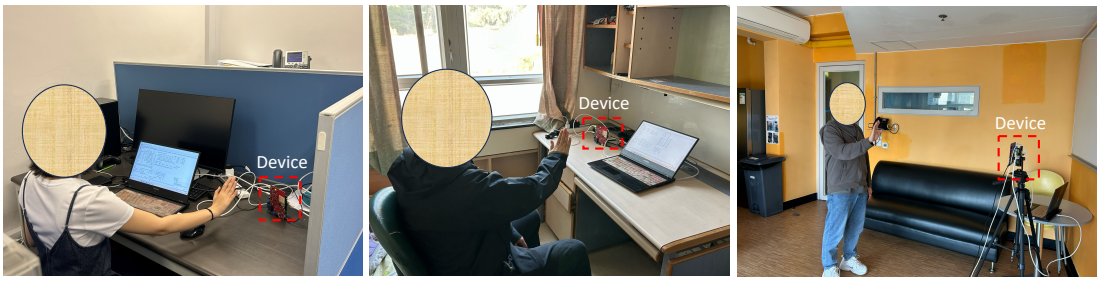
This design offers multiple advantages by decoupling the encoder from task-specific classifiers, enabling the system to support a variety of tasks without requiring extensive redesign or retraining, thereby enhancing flexibility and scalability. The lightweight nature of the encoder and task-specific modules ensures resource efficiency, making the system suitable for deployment in resource-constrained environments. Additionally, allowing partial fine-tuning of the encoder when necessary improves convergence by facilitating better adaptation to new tasks and capturing task-specific nuances effectively. Compared to methods that freeze large encoders or require significant retraining, this design strikes a balance between leveraging pretrained knowledge and maintaining adaptability, ensuring the system evolves alongside changing user demands and enhances its real-world applicability.

5 EVALUATION

In this section, we introduce the experiment setup and evaluate SEGALL to verify its effectiveness.

5.1 Experiment Setup

Signal implementation For mmWave implementation, we adopt the TI AWR2944 EVM millimeter-wave radar module, which features 4 transmit and 4 receive antennas and operates at 77 GHz. The radar is connected to the DCA1000EVM data capture board to acquire raw intermediate frequency (IF) signals, which are then transmitted to a laptop via Ethernet. Key parameters include a frame time of $80ms$, chirp duration of $625\mu s$, and a 160° FOV,



(a) Office. (b) Bedroom. (c) Living room.

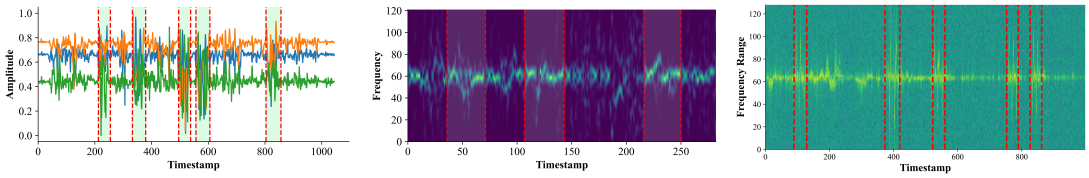
Fig. 9. Experimental environments for mmWave data collection.

suitable for required resolution of indoor scenarios to avoid excessive data size. The radar is deployed in three distinct environments—office, bedroom, and living room—as illustrated in Fig. 9. The living room is the most spacious and open, the office is relatively small with moderate obstacles, and the bedroom contains the most clutter, including bookshelves and wardrobes. The experiment involves six volunteers (three males and three female) with diverse body types in terms of weight (47-85kg) and height (158-182cm). For each setting, six sessions of continuous 30-minute data are recorded by each volunteer. Volunteers perform various daily activities (e.g., walking, typing, drinking water) around the device. At random time points, they face the device and execute target gestures that reflect real-world spontaneity. We collect six fine-grained gestures for downstream evaluation: push hand, pull hand, slide hand, wave hand, clap hand, and make a fist.

For IMU implementation, we utilize the inertial sensors embedded in an iPhone 14 at a sampling rate of 100 Hz to record long-duration human motion time series. All raw IMU data is subsequently calibrated, normalized, and downsampled to ensure consistency and reduce sensor noise. The experiment involves same six volunteers, performing gestures in the three environments as the mmWave setup: office, bedroom, and living room. Under each environment, we collect six consecutive 30-minute data sessions per user. During data collection, each participant holds the phone in hand and performs the target gestures, including sweep hand, clap hand, slide hand, wave hand, draw-O by hand, and draw-Z by hand. In between gesture executions, a range of natural daily activities—such as walking and interacting with light switches—are performed randomly and treated as interference.

For Wi-Fi implementation, we adopt a gesture-level synthesis approach using real human activities as interference. Specifically, based on the public Widar3.0 dataset [58], a Wi-Fi sensing gesture recognition dataset with more than 200,000 samples, which includes more than 10 gesture classes performed by 16 volunteers (12 males, 4 females) with diverse body shapes (height: 155–185 cm; weight: 44–89 kg) in three different indoor environments. We select 6 gestures (push&pull, sweep, clap, slide, draw-Z, draw-O) as target gestures. The remaining gestures (e.g., draw-triangle, draw-rectangle, draw-N) are randomly inserted between target gestures as interference activities. Both the number and type of interference gestures between target actions are randomly sampled to construct continuous, variable-length sequences. This synthesis strategy preserves the semantic and temporal complexity of human motion and forces the model to distinguish between subtle gesture differences, rather than simply identifying transitions between structured data segments or noise insertions. The synthetic sequences is able to reflect real-world sensing conditions, where gestures are embedded in continuous streams of diverse and ambiguous human activities.

During the data collection, the device is kept to run and record. To enable efficient and scalable ground truth annotation, we develop a lightweight Android app that allows volunteers to self-annotate gestures in real time by recording timestamps at the start and end of each target action. During each 30-minute session, volunteers



(a) IMU (Accelerometer). (b) Wi-Fi. (c) mmWave.

Fig. 10. The data demonstration for IMU (Accelerometer as example), Wi-Fi, and mmWave. The red dashed box represents the target activity.

perform daily activities including walking, typing keyboard, drinking water, and etc. Only in random time points, volunteers will tap the app to mark gesture boundaries and conduct target gestures. Each minute session typically contains 0-10 target gestures, depending on the volunteer's pace. The annotation per gesture takes less than 4 seconds. A synchronized start ensures alignment between sensing data and annotations, and a post-processing script automatically generates segmentation labels from the recorded timestamps. This semi-automated approach significantly reduces annotation overhead while maintaining high labeling accuracy across multiple environments.

Training Configuration In the training setup, the proposed algorithms are implemented using the PyTorch framework and executed on a desktop equipped with an NVIDIA A4000 GPU. The Transformer layer is configured with eight attention heads, two encoder layers, and a hidden embedding dimensionality of 128. Subsequently, a single-layer LSTM network is employed to process the outputs of the Transformer. The active learning process consists of 20 iterations, with three samples selected for annotation in each iteration. Each training phase span 10 epochs per iteration. Model optimization is conducted using the Adam optimizer, with the learning rate set to 0.001.

Baseline To validate the segmentation performance of our proposed framework SEGALL, we incorporate three representative baseline methods from the literature: a threshold-based method, a CPD-based method and a deep-learning-based approach. All baseline models are adapted to our unified data format and trained using the same ground truth annotation scheme for fair comparison. CPD-based [3] method is a change point detection approach that segments sensing data by computing a kernel-based separation distance between consecutive fixed-length change point views. It estimates the ratio of probability densities before and after each candidate point via Gaussian kernels, and designates a point as an activity transition when its separation score exceeds a threshold and forms a local peak. DeepSeg [42] is CNN-based method that classifies fixed-length bins into four states (static, start, motion, end) to detect activity boundaries. It relies on state transitions and a feedback mechanism to refine segmentation, but assumes clear gaps between activities, which limits its applicability in dense activity streams. A threshold-based method [15] is also included for comparison, as discussed in Sec. 2. All baselines are implemented or re-implemented to align with our experimental setup.

Evaluation Metric Since the segmentation model outputs a probability sequence ranging between $[0, 1]$, where only a value of 1 represents the target activity, we require a metric to evaluate the effectiveness of SEGALL. A probability value is set to 1 only when the confidence exceeds 0.7; otherwise, it is set to 0, indicating either no activity or interference activity. The resulting sequence is then aligned with the ground truth. The transition from 0 to 1 marks the start of a target activity, while the transition from 1 to 0 signifies the end of a target activity. To evaluate each detected target activity, we reference the IoU (Intersection over Union) metric [36]. IoU is calculated as:

$$\text{IoU} = \frac{\text{Duration of Overlap}}{\text{Duration of Union}}$$

A segment is considered correct only if the IoU value exceeds 0.85. However, the model may mistakenly classify interference activities as target activities. To calculate the final accuracy, we use the following formula:

$$\text{Accuracy} = \frac{\text{Correctly Detected Segments}}{\text{Total Number of Detected Segments}}$$

The total number of detected segments includes: 1. Correctly detected segments (IoU $\geq$ 0.85), 2. Incorrectly detected segments (IoU $<$ 0.85), 3. Misclassified interference activities.

5.2 Effectiveness and Universality for AL-based Segmentation

We evaluate the effectiveness and generalizability of SEGALL for segmentation tasks across three distinct wireless signal modalities: IMU, Wi-Fi, and mmWave. As shown in Fig.11, we compare SEGALL against three representative baselines: a threshold-based method [15], a CPD-based method [3], and a CNN-based DeepSeg approach [42], all of which have been discussed. The results demonstrate that SEGALL achieves an impressive average accuracy of 93.15% across these modalities, significantly outperforming baselines (only 56.43% average accuracy). Specifically, the accuracy of SEGALL for IMU, Wi-Fi, and mmWave signals is 95.10%, 93.23%, and 91.12%, respectively. Empowering by active learning, the average labeling effort is reduced by 40.15% compared to normal training, as displayed in Fig. 12. Specifically, the reduction in labeling effort for IMU, Wi-Fi, and mmWave signals is 42.33%, 40.67%, and 37.45%, respectively.

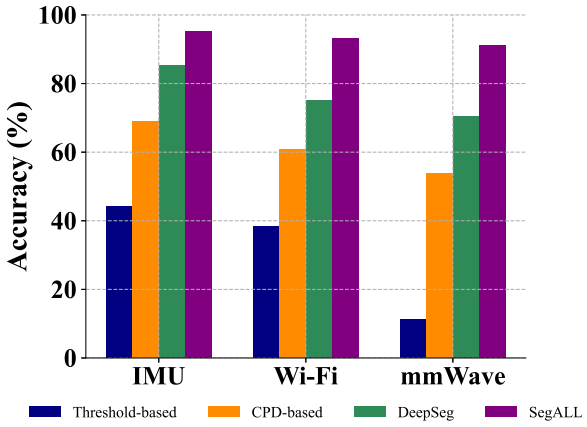


Fig. 11. Segmentation performance across various signals.

The high accuracy highlights the robustness and adaptability of SEGALL when applied to diverse wireless signals under interference. The relatively higher accuracy for IMU signals can be attributed to their lower complexity and higher signal stability. In contrast, Wi-Fi and mmWave signals involve increased complexity due to their higher resolution and susceptibility to environmental factors. This complexity increases the challenge of accurate signal interpretation, which is reflected in the slight drop in accuracy for these modalities. As for threshold-based segmentation, it struggles to process mmWave data, achieving only around 11.21% accuracy due to challenges in real-world data collection. It fails to handle consecutive interference and target activities effectively. While its performance improves on IMU and Wi-Fi data, achieving accuracies of 44.10% and 38.95%, respectively, due to the distinct distributions of real data and noise that are easier for threshold-based methods to capture, it remains significantly inferior to SEGALL. CPD-based segmentation detects change points by comparing local

kernel-based statistics across time windows. It improves upon the threshold method, achieving 69.02%, 60.67% , and 53.78% across three signals. However, it is still limited by its dependence on predefined window sizes and its sensitivity to parameter tuning, which impacts robustness across different signal domains. Compared with traditional segmentation methods, AI-based DeepSeg achieves significantly higher performance across all modalities among the baselines. Specifically, DeepSeg attains accuracies of 85.33%, 74.95%, and 70.50% on IMU, Wi-Fi, and mmWave data, respectively. This improvement largely stems from the application of deep learning, which enables an end-to-end segmentation pipeline that automatically learns to locate activity boundaries, rather than relying on manually designed thresholding rules. These results highlight the effectiveness and strength of deep learning models in addressing complex segmentation tasks. However, DeepSeg assumes clear temporal separation between activities and lacks the design for interference activities, which limit its applicability in continuous gesture streams characterized by overlapping transitions. Consequently, despite its strong performance, it still lags behind SEGALL, which is specifically designed to handle more challenging scenarios.

Overall, SEGALL consistently delivers high accuracy across all three modalities, showcasing its effectiveness in handling wireless signals of varying resolutions and complexities. Furthermore, the results highlight its robust ability to resist interference, whether originating from real-world environments or synthetic noise.

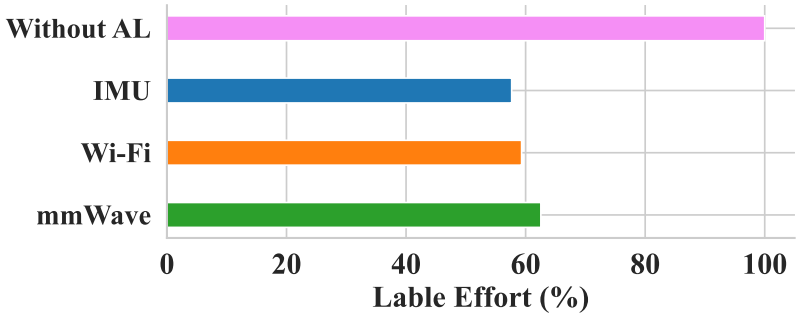


Fig. 12. Reduced label effort after applying active learning across signals.

By integrating active learning, SEGALL effectively identifies and selects the most informative samples from the unlabeled data pool, thereby avoiding a large amount of redundant labeling. Our tailored query strategy prioritizes the training across all three signals by at least 37.45% reducing labeling. Among them, the improvement based on active learning on IMU and Wi-Fi outperforms than mmWave. On the one hand, the lower complexity of IMU and Wi-Fi data alleviates required labeling efforts under proposed powerful query strategy. On the other hand, Wi-Fi is originally collected in relatively simple experiment settings although we add extra noise to simulate a more challenging scenario, still containing numerous repetitive samples. In contrast, the self-collected mmWave dataset is subject to greater interference from practical daily activities, resulting in increased data variability and a slight degradation in segmentation accuracy. Despite these challenges, the reduction in labeling effort for mmWave signals still represents a significant improvement to users.

In summary, the combination of high segmentation accuracy, adaptability to diverse wireless signals, and reduced labeling effort through active learning demonstrates the effectiveness and universality of SEGALL for AL-based segmentation tasks.

5.3 Downstream Task Performance

We further evaluate the performance of SEGALL on a downstream task to assess its practical applicability across different signal modalities. For this analysis, we select gesture recognition as an example, a relevant and challenging task in smart home scenarios where gesture control serves as a key interaction modality for various devices.

After the segmentation phase, SEGALL processes the well-segmented data to recognize gestures, as illustrated in Fig. 13. The results show that SEGALL achieves high recognition accuracy across all three signal types, with 97.15% accuracy for IMU, 91.67% for Wi-Fi, and 92.75% for mmWave signals. These results highlight the ability of SEGALL to maintain robust performance in downstream tasks, regardless of the signal type. Similar to the segmentation performance, IMU signals achieve the highest recognition accuracy due to their simplicity and stability. It is important to note that the mmWave signal is expected to achieve higher recognition accuracy than Wi-Fi due to its superior resolution. However, in our self-collected dataset, the mmWave gestures are performed casually across three rooms rather than in a standardized manner, which introduces variability and reduces performance. This factor explains why the performance of mmWave and Wi-Fi signals is comparable in this case.

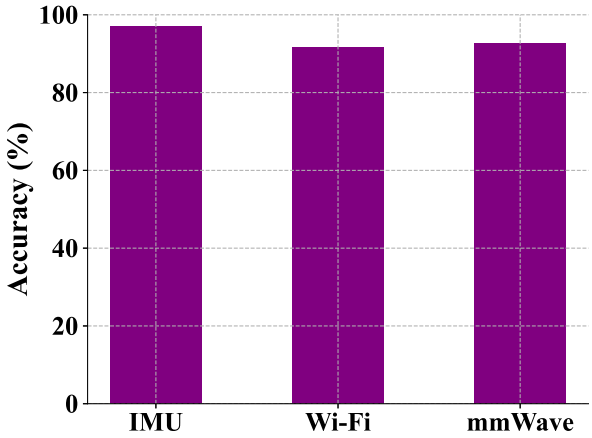


Fig. 13. Downstream task performance (gesture recognition as an example).

Overall, this evaluation demonstrates that SEGALL is not only capable of accurately segmenting data but also excels in downstream tasks such as gesture recognition across different wireless signals.

5.4 Real-time Analysis

We analyze the real-time inference performance of SEGALL on various platforms to demonstrate its capability to operate in real-time scenarios. Table 3 presents the processing latency (in milliseconds) for different modalities, including IMU, Wi-Fi, and mmWave, across three platforms: PC (RTX 4090), Macbook (M2 Pro), and Raspberry Pi. To ensure a fair comparison, all experiments use a processing window size of 1 minute with a sampling rate of 15 Hz.

SEGALL demonstrates remarkable real-time performance, especially on high-performance platforms like PC and Macbook. For the IMU modality, the latency on a PC is only 1.80 ms, significantly below the real-time threshold. Even on the Macbook, the latency remains at 23.31 ms, which is well-suited for real-time applications. The Raspberry Pi, being a resource-constrained edge device, still achieves acceptable latency for IMU at 150.21 ms, demonstrating the system's adaptability across diverse platforms. For Wi-Fi and mmWave modalities, which involve more complex signal processing, SEGALL maintains competitive performance. The latency on a PC for

Table 3. Real-time inference performance (ms) of SEGALL within different platforms. For fair comparison, the processing window size for each modality is 1 minute and sampling rate is 15 Hz.

Configuration	PC	Macbook	Raspberry Pi
IMU	1.80	23.31	150.21
Wi-Fi	10.98	37.05	463.15
mmWave	9.75	71.52	632.35

Wi-Fi is 10.98 ms and for mmWave is 9.75 ms, both of which are efficient for real-time processing. On a Macbook, while the latency increases to 37.05 ms for Wi-Fi and 71.52 ms for mmWave, these values are still within the bounds of real-time usability. On Raspberry Pi, the latency for Wi-Fi and mmWave increases further to 463.15 ms and 632.35 ms, respectively, reflecting the computational limitations of the platform. However, this performance is still valuable for scenarios where ultra-low latency is not critical.

Overall, SEGALL achieves a balance between computational efficiency and real-time performance across different platforms and modalities. The results highlight the system's capability to operate effectively in diverse environments, from high-performance computing to resource-constrained edge devices, ensuring its applicability to a wide range of real-time applications.

5.5 Convergence Efficiency of Reusable Module

We analyze the training convergence speed of SEGALL during the training process for the downstream task, focusing on the role of the common encoder. Initially, we experiment with freezing the encoder and fine-tuning only the task classifier at the beginning of the training process. However, this approach faces challenges in achieving convergence as illustrated in Fig. 14. The primary reasons for this difficulty are the mismatch in objectives between the pretrained encoder and the downstream task, as well as the limited number of parameters available for fine-tuning. SEGALL adopts a strategy of training all parameters, including the encoder, based on the existing pretrained encoder. This approach allows the system to leverage the encoder's foundational knowledge while adapting it more effectively to the nuances of the downstream task. Proposed strategy accelerates the convergence process. Specifically, the number of epochs required for convergence is reduced to 15 compared to 60 epochs with retraining a new encoder from scratch. By jointly optimizing all parameters, SEGALL achieves a balance between generalization and task-specific specialization, ensuring both efficiency and performance.

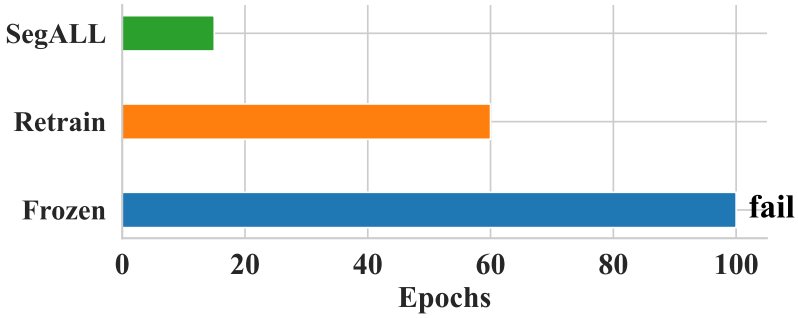


Fig. 14. Comparison of converge efficiency.

5.6 Ablation Analysis

For verifying the effectiveness and necessity of SEGALL designs, we perform ablation studies from different perspectives.

Comparison of Different Query Methods. We compare the proposed Dispersion Metric with previous active learning query strategies to investigate their impact on reducing manual labeling efforts during the segmentation training process, as shown in Fig. 15. The effectiveness of each query strategy is measured using the number of required manually labeled data points to achieve final convergence.

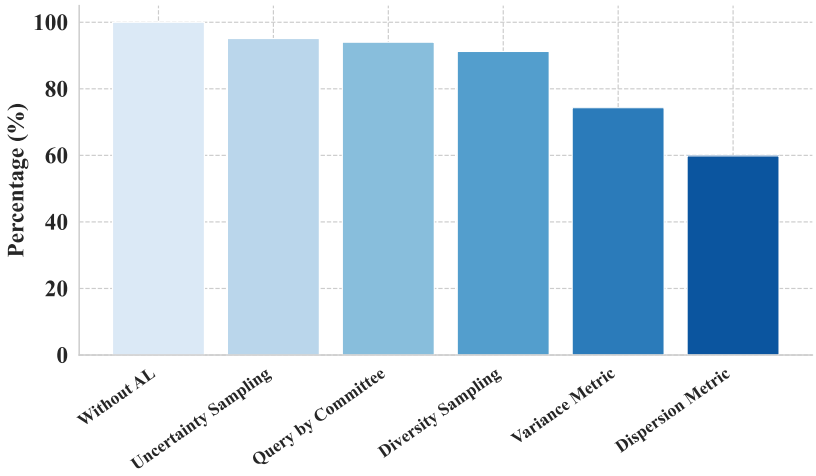


Fig. 15. Labeled efforts can be reduced by different query strategies.

Traditional active learning methods are primarily designed for classification problems, where the focus is on selecting the most uncertain samples for labeling. However, these methods are less effective in segmentation tasks, which require measuring general information from a sequence of data rather than individual samples. This limitation makes it challenging for these methods to identify the most informative samples, especially in scenarios where the data series contain significant interference or noise. The proposed basic variance metric reduces manual labeling efforts by 25.67% compared to previous methods, which still perform at a similar level to traditional training with no active selection mechanism. Building upon the basic variance metric, the proposed Dispersion Metric effectively identifies the most challenging and representative samples for labeling. As a result, the Dispersion Metric reduces manual labeling efforts by an impressive 40.15%, a substantial improvement over both traditional active learning methods and the basic variance metric. These results highlight the effectiveness of the Dispersion Metric in scenarios with limited labeling data.

Comparison of Different Model Backbones. We design a tailored model backbone for SEGALL, as discussed in Sec 4.2. In this part, we aim to examine the effectiveness of this architecture by replacing it with single component. Specifically, we consider a single Transformer backbone and a single LSTM backbone as alternative reused encoders to compare their performance with our design.

As shown in Fig. 16, during the segmentation phase, the Transformer backbone achieves an average accuracy of only 61.09% across the three signal modalities. Its performance decreases further as the complexity of the modality increases. On the other hand, the single LSTM backbone performs even worse, achieving an average accuracy of only 36.12%. The poor performance of the LSTM backbone is primarily due to its simplicity, which limits its

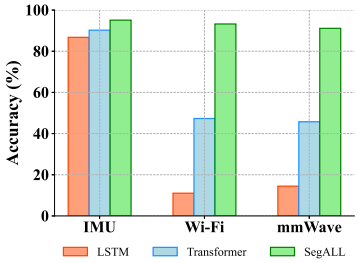


Fig. 16. Performance with different backbones during the segmentation phase.

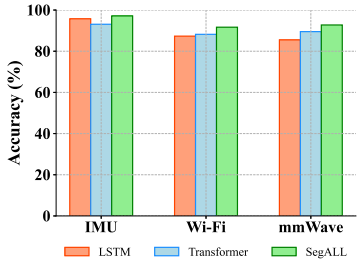


Fig. 17. Performance with different backbones during the downstream task phase.

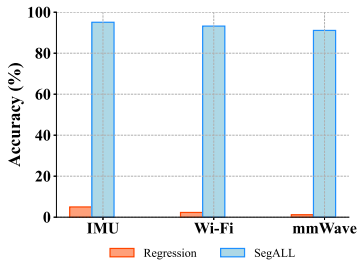


Fig. 18. Performance with different segmentation methods.

ability to extract effective features from complex, high-dimensional data. SEGALL integrates both Transformer and LSTM architectures, leveraging their complementary strengths: the Transformer excels at capturing long-range dependencies and global temporal patterns, while the LSTM is better suited for modeling fine-grained sequential transitions. This hybrid design enables SEGALL to achieve a more powerful and generalizable segmentation. For the downstream task shown in Fig. 17, both Transformer and LSTM backbones demonstrate acceptable accuracy (above 85%) when applied to well-segmented data from different signal types. This further explains the key to solve practical deployment problem is accurate segmentation. Therefore, SEGALL has the better architecture to deal with segmentation and recognition challenges.

Comparison of Different Segmentation Methods. We investigate the rationality of treating the segmentation method as a binary classification problem rather than a regression problem, particularly in the context of long time-series data where target activities are sparse.

As shown in Fig. 18, treating segmentation as a regression problem—where the model directly outputs segmentation points—results in a significant accuracy drop, with performance dropping to nearly 0% accuracy. This failure proves it is extremely challenging for a regression-based approach to precisely predict segmentation points in such sparse settings. In contrast, binary classification task outputs probabilities to determine a 0-1 sequence representing the presence or absence of target activities—greatly solve this issue and ensures a more stable and accurate segmentation performance.

6 DISCUSSION

In this section, we discuss several limitations of SEGALL and outline directions for future work.

Multi-person Detection. In this work, we primarily focus on segmenting and recognizing the activities of a single person. This approach is sufficient for many controlled scenarios, such as laboratory experiments and smart-home environments. For example, in smart-home settings, a single user often controls appliances like turning them on or off, making single-person activity detection adequate for most use cases. However, scenarios involving multiple people introduce significant challenges. When multiple individuals are present, separating and accurately recognizing their respective activities becomes far more complex. This complexity arises from the need to handle multi-target signal separation, which each modality has its unique characteristics and processing requirements, making multi-person detection a non-trivial task. Moreover, incorporating additional computational steps for multi-target processing could result in delays, necessitating efficient algorithms and hardware optimization. Addressing these challenges—both in terms of signal processing and real-time performance—is a key direction for our future work.

System Deployment. Another important consideration is the deployment of the SEGALL algorithm onto resource-constrained devices, such as embedded chips in IoT systems. In the current work, we evaluate the performance of SEGALL on a standard computer platform and process a fixed-length segment of data after collection. This approach is primarily due to the lack of access to programmable commercial radar equipment and the limitations imposed by closed-source hardware. Notably, the TI mmWave radar used in our system does not provide an interface for continuous data streaming, which restricts SEGALL's ability to support true streaming or overlapping-window processing at this stage. To enable real-world adoption, it is crucial to implement the SEGALL algorithm on embedded systems, such as edge devices or IoT hardware. Achieving this goal would require close collaboration with hardware manufacturers to optimize the algorithm for deployment on low-power, programmable chips. Techniques such as model compression, quantization, and efficient signal processing pipelines can play a pivotal role in enabling lightweight and real-time execution of SEGALL on edge devices. Furthermore, we consider the integration of streaming and overlapping-window processing as promising directions for future work, especially as the hardware becomes more flexible and programmable.

7 CONCLUSION

In this paper, we proposed SEGALL, a comprehensive active learning-based framework for segmenting long, continuous wireless sensing data. SEGALL overcomes key challenges in wireless sensing, including handling interference, addressing signal complexity, and reducing labeling efforts. The results highlight SEGALL's potential to revolutionize applications in smart homes, where efficient and accurate data processing is critical.

8 ACKNOWLEDGMENTS

We sincerely thank all the reviewers for their insightful feedback to improve this paper. This work was supported in part by National Key R&D Program Project No. 2024YFE0203700, ECS CityU 21216822, CRF C1045-23G, CityU 11206023, NSFC No. 62272317, and Shenzhen Science and Technology program No. JCYJ20220531103402006.

REFERENCES

[1] Heba Abdelnasser, Khaled Harras, and Moustafa Youssef. A ubiquitous wifi-based fine-grained gesture recognition system. IEEE Transactions on Mobile Computing, 18(11):2474–2487, 2018.

[2] Mohammad Arif Ul Alam, Nirmalya Roy, Aryya Gangopadhyay, and Elizabeth Galik. A smart segmentation technique towards improved infrequent non-speech gestural activity recognition model. Pervasive and Mobile Computing, 34:25–45, 2017.

[3] Samaneh Aminikhanghahi and Diane J Cook. Enhancing activity recognition using cpd-based activity segmentation. Pervasive and Mobile Computing, 53:75–89, 2019.

[4] Yiming Bao, Xu Zhao, and Dahong Qian. Fusepose: Imu-vision sensor fusion in kinematic space for parametric human pose estimation. IEEE Transactions on Multimedia, 25:7736–7746, 2022.

[5] Qiming Cao, Hongfei Xue, Tianci Liu, Xingchen Wang, Haoyu Wang, Xincheng Zhang, and Lu Su. mmclip: Boosting mmwave-based zero-shot har via signal-text alignment. In Proceedings of the 22nd ACM Conference on Embedded Networked Sensor Systems, pages 184–197, 2024.

[6] Zhaoxin Chang, Fusang Zhang, Jie Xiong, Junqi Ma, Beihong Jin, and Daqing Zhang. Sensor-free soil moisture sensing using lora signals. Proceedings of the ACM on Interactive, Mobile, Wearable and Ubiquitous Technologies, 6(2):1–27, 2022.

[7] Guoxuan Chi, Zheng Yang, Chenshu Wu, Jingao Xu, Yuchong Gao, Yunhao Liu, and Tony Xiao Han. Rf-diffusion: Radio signal generation via time-frequency diffusion. In Proceedings of the 30th Annual International Conference on Mobile Computing and Networking, pages 77–92, 2024.

[8] Ruiyang Gao, Wenwei Li, Jinyi Liu, Shuyu Dai, Mi Zhang, Leye Wang, and Daqing Zhang. Wicgesture: Meta-motion-based continuous gesture recognition with wi-fi. IEEE Internet of Things Journal, 11(9):15087–15099, 2023.

[9] Ruiyang Gao, Mi Zhang, Jie Zhang, Yang Li, Enze Yi, Dan Wu, Leye Wang, and Daqing Zhang. Towards position-independent sensing for gesture recognition with wi-fi. Proceedings of the ACM on Interactive, Mobile, Wearable and Ubiquitous Technologies, 5(2):1–28, 2021.

[10] Wenfeng He, Kaishun Wu, Yongpan Zou, and Zhong Ming. Wig: Wifi-based gesture recognition system. In 2015 24th International Conference on Computer Communication and Networks (ICCCN), pages 1–7. IEEE, 2015.

[11] Yinghui He, Jianwei Liu, Mo Li, Guanding Yu, Jinsong Han, and Kui Ren. Sencom: Integrated sensing and communication with practical wifi. In Proceedings of the 29th Annual International Conference on Mobile Computing and Networking, pages 1–16, 2023.
[12] Hideitsu Hino and Shinto Eguchi. Active learning by query by committee with robust divergences. Information Geometry, 6(1):81–106, 2023.
[13] Yuqian Hu, Feng Zhang, Chenshu Wu, Beibei Wang, and KJ Ray Liu. Defall: Environment-independent passive fall detection using wifi. IEEE Internet of Things Journal, 9(11):8515–8530, 2021.
[14] Dheryta Jaisinghani, Rajesh Krishna Balan, Vinayak Naik, Archan Misra, and Youngki Lee. Experiences & challenges with server-side wifi indoor localization using existing infrastructure. In Proceedings of the 15th EAI International Conference on Mobile and Ubiquitous Systems: Computing, Networking and Services, pages 226–235, 2018.
[15] Robert M Keenan and Le-Nam Tran. Fall detection using wi-fi signals and threshold-based activity segmentation. In 2020 IEEE 31st Annual International Symposium on Personal, Indoor and Mobile Radio Communications, pages 1–6. IEEE, 2020.
[16] Abdelwahed Khamis, Branislav Kusy, Chun Tung Chou, Mary-Louise McLaws, and Wen Hu. Rfwash: a weakly supervised tracking of hand hygiene technique. In Proceedings of the 18th conference on embedded networked sensor systems, pages 572–584, 2020.
[17] Ka-Ho Lam, Chi-Chung Cheung, and Wah-Ching Lee. Rssi-based lora localization systems for large-scale indoor and outdoor environments. IEEE Transactions on Vehicular Technology, 68(12):11778–11791, 2019.
[18] Yong Jae Lee, Jaechul Kim, and Kristen Grauman. Key-segments for video object segmentation. In 2011 International conference on computer vision, pages 1995–2002. IEEE, 2011.
[19] Fuxin Li, Taeyoung Kim, Ahmad Humayun, David Tsai, and James M Rehg. Video segmentation by tracking many figure-ground segments. In Proceedings of the IEEE international conference on computer vision, pages 2192–2199, 2013.
[20] Hang Li, Xi Chen, Ju Wang, Di Wu, and Xue Liu. Dafi: Wifi-based device-free indoor localization via domain adaptation. Proceedings of the ACM on Interactive, Mobile, Wearable and Ubiquitous Technologies, 5(4):1–21, 2021.
[21] Yanwei Li, Hengshuang Zhao, Xiaojuan Qi, Liwei Wang, Zeming Li, Jian Sun, and Jiaya Jia. Fully convolutional networks for panoptic segmentation. In Proceedings of the IEEE/CVF conference on computer vision and pattern recognition, pages 214–223, 2021.
[22] Feng Lin, Yan Zhuang, Chen Song, Aosen Wang, Yiran Li, Changzhan Gu, Changzhi Li, and Wenyao Xu. Sleepsense: A noncontact and cost-effective sleep monitoring system. IEEE transactions on biomedical circuits and systems, 11(1):189–202, 2016.
[23] Jinpeng Lin, Zhihao Liang, Shengheng Deng, Lile Cai, Tao Jiang, Tianrui Li, Kui Jia, and Xun Xu. Exploring diversity-based active learning for 3d object detection in autonomous driving. IEEE Transactions on Intelligent Transportation Systems, 2024.
[24] T Lin. Focal loss for dense object detection. arXiv preprint arXiv:1708.02002, 2017.
[25] Haipeng Liu, Kening Cui, Kaiyuan Hu, Yuheng Wang, Anfu Zhou, Liang Liu, and Huadong Ma. mtranssee: Enabling environment-independent mmwave sensing based gesture recognition via transfer learning. Proceedings of the ACM on Interactive, Mobile, Wearable and Ubiquitous Technologies, 6(1):1–28, 2022.
[26] Xuefeng Liu, Jiannong Cao, Shaojie Tang, and Jiaqi Wen. Wi-sleep: Contactless sleep monitoring via wifi signals. In 2014 IEEE Real-Time Systems Symposium, pages 346–355. IEEE, 2014.
[27] Jonathan Long, Evan Shelhamer, and Trevor Darrell. Fully convolutional networks for semantic segmentation. In Proceedings of the IEEE conference on computer vision and pattern recognition, pages 3431–3440, 2015.
[28] Yongsen Ma, Gang Zhou, Shuangquan Wang, Hongyang Zhao, and Woosub Jung. Signfi: Sign language recognition using wifi. Proceedings of the ACM on Interactive, Mobile, Wearable and Ubiquitous Technologies, 2(1):1–21, 2018.
[29] Prateek Munjal, Nasir Hayat, Munawar Hayat, Jamshid Sourati, and Shadab Khan. Towards robust and reproducible active learning using neural networks. In Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition, pages 223–232, 2022.
[30] Armando Muscariello, Guillaume Gravier, and Frédéric Bimbot. Variability tolerant audio motif discovery. In Advances in Multimedia Modeling: 15th International Multimedia Modeling Conference, MMM 2009, Sophia-Antipolis, France, January 7-9, 2009. Proceedings. 15, pages 275–286. Springer, 2009.
[31] Qin Ni, Timothy Patterson, Ian Cleland, and Chris Nugent. Dynamic detection of window starting positions and its implementation within an activity recognition framework. Journal of biomedical informatics, 62:171–180, 2016.
[32] Kai Niu, Fusang Zhang, Xuanzhi Wang, Qin Lv, Haitong Luo, and Daqing Zhang. Understanding wifi signal frequency features for position-independent gesture sensing. IEEE Transactions on Mobile Computing, 21(11):4156–4171, 2021.
[33] Anestis Papazoglou and Vittorio Ferrari. Fast object segmentation in unconstrained video. In Proceedings of the IEEE international conference on computer vision, pages 1777–1784, 2013.
[34] Alwin Poulose, Jihun Kim, and Dong Seog Han. A sensor fusion framework for indoor localization using smartphone sensors and wi-fi rssi measurements. Applied Sciences, 9(20):4379, 2019.
[35] Pengzhen Ren, Yun Xiao, Xiaojun Chang, Po-Yao Huang, Zhihui Li, Brij B Gupta, Xiaojiang Chen, and Xin Wang. A survey of deep active learning. ACM computing surveys (CSUR), 54(9):1–40, 2021.
[36] Hamid Rezatofighi, Nathan Tsoi, JunYoung Gwak, Amir Sadeghian, Ian Reid, and Silvio Savarese. Generalized intersection over union: A metric and a loss for bounding box regression. In Proceedings of the IEEE/CVF conference on computer vision and pattern recognition, pages 658–666, 2019.

[37] Burr Settles. Active learning literature survey. 2009.
[38] Hao Wang, Daqing Zhang, Yasha Wang, Junyi Ma, Yuxiang Wang, and Shengjie Li. Rt-fall: A real-time and contactless fall detection system with commodity wifi devices. IEEE Transactions on Mobile Computing, 16(2):511–526, 2016.
[39] Haochen Wang, Xiaolong Jiang, Haibing Ren, Yao Hu, and Song Bai. Swiftnet: Real-time video object segmentation. In Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition, pages 1296–1305, 2021.
[40] Wei Wang, Alex X Liu, Muhammad Shahzad, Kang Ling, and Sanglu Lu. Understanding and modeling of wifi signal based human activity recognition. In Proceedings of the 21st annual international conference on mobile computing and networking, pages 65–76, 2015.
[41] Dan Wu, Youwei Zeng, Fusang Zhang, and Daqing Zhang. Wifi csi-based device-free sensing: from fresnel zone model to csi-ratio model. CCF Transactions on Pervasive Computing and Interaction, pages 1–15, 2022.
[42] Chunjing Xiao, Yue Lei, Yongsen Ma, Fan Zhou, and Zhiguang Qin. Deepseg: Deep-learning-based activity segmentation framework for activity recognition using wifi. IEEE Internet of Things Journal, 8(7):5669–5681, 2020.
[43] Binbin Xie, Yuqing Yin, and Jie Xiong. Pushing the limits of long range wireless sensing with lora. Proceedings of the ACM on Interactive, Mobile, Wearable and Ubiquitous Technologies, 5(3):1–21, 2021.
[44] Huatao Xu, Pengfei Zhou, Rui Tan, Mo Li, and Guobin Shen. Limu-bert: Unleashing the potential of unlabeled data for imu sensing applications. In Proceedings of the 19th ACM Conference on Embedded Networked Sensor Systems, pages 220–233, 2021.
[45] Yu-Syuan Xu, Tsu-Jui Fu, Hsuan-Kung Yang, and Chun-Yi Lee. Dynamic video segmentation network. In Proceedings of the IEEE conference on computer vision and pattern recognition, pages 6556–6565, 2018.
[46] Hongfei Xue, Qiming Cao, Chenglin Miao, Yan Ju, Haochen Hu, Aidong Zhang, and Lu Su. Towards generalized mmwave-based human pose estimation through signal augmentation. In Proceedings of the 29th Annual International Conference on Mobile Computing and Networking, pages 1–15, 2023.
[47] Weixing Xue, Weining Qiu, Xianghong Hua, and Kegen Yu. Improved wi-fi rssi measurement for indoor localization. IEEE Sensors Journal, 17(7):2224–2230, 2017.
[48] Yi Yang, Zhigang Ma, Feiping Nie, Xiaojun Chang, and Alexander G Hauptmann. Multi-class active learning by uncertainty sampling with diversity maximization. International Journal of Computer Vision, 113:113–127, 2015.
[49] Zheng Yang, Yi Zhang, and Qian Zhang. Rethinking fall detection with wi-fi. IEEE Transactions on Mobile Computing, 22(10):6126–6143, 2022.
[50] Yicheng Yao, Changyu Liu, Hao Zhang, Baiju Yan, Pu Jian, Peng Wang, Lidong Du, Xianxiang Chen, Baoshi Han, and Zhen Fang. Fall detection system using millimeter-wave radar based on neural network and information fusion. IEEE Internet of Things Journal, 9(21):21038–21050, 2022.
[51] Donggeun Yoo and In So Kweon. Learning loss for active learning. In Proceedings of the IEEE/CVF conference on computer vision and pattern recognition, pages 93–102, 2019.
[52] Youwei Zeng, Dan Wu, Jie Xiong, Enze Yi, Ruiyang Gao, and Daqing Zhang. Farsense: Pushing the range limit of wifi-based respiration sensing with csi ratio of two antennas. Proceedings of the ACM on Interactive, Mobile, Wearable and Ubiquitous Technologies, 3(3):1–26, 2019.
[53] Duo Zhang, Xusheng Zhang, Shengjie Li, Yaxiong Xie, Yang Li, Xuanzhi Wang, and Daqing Zhang. Lt-fall: The design and implementation of a life-threatening fall detection and alarming system. Proceedings of the ACM on Interactive, Mobile, Wearable and Ubiquitous Technologies, 7(1):1–24, 2023.
[54] Fusang Zhang, Zhaoxin Chang, Jie Xiong, and Daqing Zhang. Exploring lora for sensing. GetMobile: Mobile Computing and Communications, 25(2):33–37, 2021.
[55] Xi Zhang, Yu Zhang, Zhenguo Shi, and Tao Gu. mmfer: Millimetre-wave radar based facial expression recognition for multimedia iot applications. In Proceedings of the 29th Annual International Conference on Mobile Computing and Networking, pages 1–15, 2023.
[56] Guangzhi Zhao, Zhipeng Zhou, Yutao Huang, Amiya Nayak, Wei Gong, and Haoquan Zhou. Alsensing: Human activity recognition using wifi based on active learning. In 2023 IEEE Wireless Communications and Networking Conference (WCNC), pages 1–6. IEEE, 2023.
[57] Naiyu Zheng, Yuanchun Li, Shiqi Jiang, Yuanzhe Li, Rongchun Yao, Chuchu Dong, Ting Chen, Yubo Yang, Zhimeng Yin, and Yunxin Liu. Adawifi, collaborative wifi sensing for cross-environment adaptation. IEEE Transactions on Mobile Computing, 2024.
[58] Yue Zheng, Yi Zhang, Kun Qian, Guidong Zhang, Yunhao Liu, Chenshu Wu, and Zheng Yang. Zero-effort cross-domain gesture recognition with wi-fi. In Proceedings of the 17th annual international conference on mobile systems, applications, and services, pages 313–325, 2019.